

VoltVision — EV Battery Swap Analytics & Decision Intelligence

Hackathon Analysis Report

Tagline: Turning EV network data into smarter decisions.

Executive Summary

VoltVision analyzes EV battery-swapping network data to understand operational performance, customer experience, station-level differences, battery and equipment behavior, pricing and partner economics, and rider retention.

The analysis combines eight datasets covering millions of swap events, station operations, riders, batteries, support activity, geographic context, and fleet partners.

The analysis focuses on six core business questions:

1. How is network performance changing over time?
2. What factors are associated with service failures and poor customer experience?
3. Which station and geographic patterns are associated with differences in performance?
4. How do battery health and equipment characteristics relate to operational usage?
5. How do pricing and fleet-partner patterns relate to revenue and network economics?
6. What factors are associated with new riders returning to the network?

The objective is not to establish causal relationships, but to provide an evidence-based view of network performance and identify areas where operations and business teams can investigate further.

Key Outcome

VoltVision transforms raw operational data into actionable analytics that can help decision-makers monitor network health, identify stations requiring attention, understand battery and pricing patterns, and investigate factors associated with rider retention.

1. Problem Understanding

VoltRelay Energy operates an EV battery-swapping network where customers exchange depleted batteries for charged ones. As the network grows, large volumes of operational data are generated across stations, batteries, riders, pricing, and fleet partnerships.

The business challenge is to turn this data into clear evidence about how the network is performing and where operational or commercial attention may be required.

The analysis focuses on understanding:

- Network performance and changes over time.
- Service failures, queue waiting times, and customer experience.
- Differences in station and geographic performance.
- Battery health and equipment utilization.
- Pricing, discounts, revenue, and fleet-partner economics.
- Factors associated with new-rider retention.

Business Objectives

The objectives of the analysis are to:

1. Measure network performance using swap volume, completion, failure, abandonment, cancellation, and revenue metrics.
2. Identify service patterns associated with failed or abandoned swap attempts.
3. Compare station performance and identify locations requiring further investigation.
4. Examine battery health and usage patterns to support equipment monitoring.
5. Understand pricing and partner-level revenue patterns.
6. Identify customer-experience factors associated with 30-day rider retention.
7. Translate analytical findings into practical business recommendations.

Analytical Scope

The analysis covers the available data from **January 2024 through June 2025** and uses the available operational, customer, battery, station, support, geographic, and partner datasets.

The analysis is observational. Identified relationships represent associations in the available data and should not be interpreted as proof of causation.

2. Data and Analytical Approach

2.1 Data Sources

The analysis uses eight datasets representing different parts of the EV battery-swapping ecosystem:

Dataset	Purpose
Swap Events	Swap attempts, outcomes, queue waits, pricing, batteries, and revenue
Station Hourly Status	Station operating conditions and hourly availability
Riders	Rider-level customer information
Batteries	Battery health and equipment characteristics
Support Tickets	Customer complaints and support interactions
Stations	Station characteristics and location information
City Daily Context	Daily geographic and environmental context
Fleet Partners	Fleet-partner information and commercial attributes

The combined datasets provide an operational view of the network from the perspectives of customers, stations, batteries, and business partners.

2.2 Data Preparation

The raw datasets were inspected for:

- Missing values.
- Duplicate records.
- Data types and date fields.
- Invalid or inconsistent records.
- Potential test or non-production activity.
- Relationships between datasets.

The raw swap-event dataset contained **3,877,013 records**. After removing identified potential duplicate records, **3,876,961 network-level events** remained.

For production-focused operational analysis, identified test-station activity was excluded, resulting in **3,814,930 production swap events**.

2.3 Analytical Approach

The analysis followed these stages:

1. Data Understanding

Examined dataset structures, dimensions, columns, data types, and data quality.

2. Data Cleaning

Checked missing values, duplicate records, date fields, and potential non-production records.

3. Exploratory Data Analysis

Explored distributions, trends, station differences, battery characteristics, pricing patterns, and rider behavior.

4. Core Question Analysis

Analyzed the six business questions defined in the problem statement.

5. Visualization

Used time-series charts, distributions, comparisons, rankings, and relationship plots to make patterns easier to interpret.

6. Business Interpretation

Converted analytical observations into operational findings and areas for further investigation.

2.4 Analytical Period

The primary analysis covers the period from **January 2024 through June 2025**.

2.5 Important Analytical Principle

The analysis is observational. Relationships identified between variables indicate associations within the available data and do not establish that one factor directly caused another.

3. Network Performance Over Time

3.1 Overview

The first analysis evaluates how the EV battery-swapping network performed over time using swap volume, completion, failure, abandonment, cancellation, revenue, and contribution-margin metrics.

The analysis period covers **January 2024 to June 2025**.

3.2 Overall Network Metrics

Metric	Result
Network-level swap attempts	3,876,961
Completed swaps	3,645,762
Completion rate	94.04%
Failure rate	3.47%
Abandonment rate	1.77%
Cancellation rate	0.43%
Total revenue	₹23.31 crore
Contribution margin per completed swap	₹44.45

3.3 Performance Analysis

Monthly swap activity was analyzed to understand changes in network demand and operational throughput.

The analysis tracked:

- Total swap attempts.
- Completed swaps.
- Completion rate.
- Failure rate.
- Abandonment rate.
- Cancellation rate.
- Monthly revenue.

These measures provide a combined view of network demand and service performance rather than relying on a single KPI.

3.4 Key Observations

The network processed millions of swap attempts during the analysis period, with the large majority resulting in completed swaps.

Failures, abandoned queues, and cancellations represent separate components of unsuccessful customer journeys and were therefore analyzed independently.

Revenue was also examined over time alongside swap activity to understand the relationship between network usage and commercial performance.

An important observation is that the observed contribution margin per swap does not necessarily follow the directional assumption described in the problem context. The analysis therefore reports the measured data rather than forcing the results to match the initial hypothesis.

3.5 Business Implication

Network leaders can use these metrics as a recurring operational scorecard. Monitoring completion, failure, abandonment, revenue, and margin together can help distinguish changes in customer demand from changes in service performance.

The monthly trend analysis also provides a baseline for identifying periods that require deeper investigation at the station, battery, pricing, or customer-experience level.

4. Service Failures and Customer Experience

4.1 Overview

The second analysis examines unsuccessful customer experiences across the network, focusing on service failures, queue waiting time, abandoned attempts, cancellations, station-level differences, and hourly patterns.

4.2 Failure and Customer Experience Analysis

Swap events were categorized into completed, failed, abandoned, and cancelled outcomes.

Failures included:

- Failure due to no charged battery.
- System-related failure.

Abandoned attempts represent customers leaving the queue before completing the swap, while cancellations represent rider-initiated cancellations.

Queue waiting time was analyzed using mean, median, P90, P95, and P99 values to understand both typical and extreme waiting experiences.

4.3 Station-Level Analysis

Station-level failure rates were calculated to identify locations with different service-performance patterns.

To reduce distortion from very low-volume stations, station comparisons focused on locations with at least **1,000 swap attempts**.

The analysis compared:

- Total attempts.
- Failed attempts.
- Failure rate.
- Abandonment rate.
- Average queue waiting time.

This approach helps distinguish consistently observed station-level patterns from results driven by very small samples.

4.4 Hourly Analysis

Service performance was also analyzed by hour of day.

Hourly metrics included:

- Swap attempts.
- Failure rate.
- Abandonment rate.
- Average queue waiting time.

This helps identify periods where demand and operational pressure may coincide with changes in customer experience.

4.5 Business Implications

Stations with relatively high failure rates, high abandonment, or long waiting times can be prioritized for operational investigation.

Hourly patterns can also support staffing, battery replenishment, and station-capacity planning.

However, these patterns are observational. A high failure rate occurring alongside longer waiting times does not by itself prove that waiting time caused the failures.

5. Station and Geographic Patterns

5.1 Overview

The third analysis examines whether operational performance differs across stations and available station characteristics.

Station-level metrics were calculated using swap volume, completion rate, failure rate, and average queue waiting time.

5.2 Station Performance

Each station was evaluated based on:

- Total swap attempts.
- Completed swaps.
- Failed attempts.
- Completion rate.
- Failure rate.
- Average queue waiting time.

Stations with at least **1,000 attempts** were used for the primary performance comparison to reduce the effect of low-volume locations.

5.3 Station Characteristics

Available station attributes were compared with operational performance to identify differences between station categories.

The analysis considered available categorical attributes such as station type, location-related characteristics, and other operational classifications present in the station dataset.

Where suitable commissioning information was available, station timing was also considered as a potential segmentation factor.

5.4 Relationship Between Waiting Time and Failures

A station-level comparison was performed between average queue waiting time and failure rate.

This helps identify whether stations experiencing longer waits also show different failure patterns.

The relationship is interpreted as an association rather than evidence that longer waiting directly causes failures.

5.5 Business Implications

Station-level analysis allows network management to move beyond overall averages and identify specific locations requiring investigation.

High-volume stations with weaker completion performance, higher failure rates, or longer queues can be reviewed alongside station characteristics and operational conditions.

This can support targeted maintenance, capacity planning, battery availability reviews, and station-level operational improvements.

6. Battery and Equipment Performance

6.1 Overview

The fourth analysis examines battery health, battery usage, charge-cycle information, and available supplier or manufacturer attributes.

The objective is to understand how battery characteristics relate to observed usage across the swapping network.

6.2 Battery Health

The battery dataset was analyzed to understand the distribution of available State of Health (SOH) and other health-related measures.

Battery health was examined alongside observed swap activity to identify differences between batteries with different levels of recorded health.

6.3 Battery Usage

Battery identifiers from swap events were matched with the battery master dataset.

For each battery, observed swap activity was calculated using its recorded involvement in incoming and outgoing swap events.

This created a battery-level view combining:

- Battery characteristics.
- Health indicators.
- Usage intensity.
- Available charge-cycle information.
- Supplier or manufacturer attributes.

6.4 Battery Health and Usage Relationship

Battery health was compared with total observed swap activity.

Batteries were also grouped into health ranges to compare their average swap activity.

This provides an operational view of whether heavily used batteries exhibit different health characteristics from less-used batteries.

6.5 Supplier and Manufacturer Analysis

Where supplier or manufacturer information was available, battery activity was compared across groups.

This can help identify differences that may warrant further investigation into battery sourcing, maintenance, or lifecycle management.

6.6 Business Implications

Battery health and usage monitoring can support preventive maintenance and fleet-management decisions.

Batteries showing unusual combinations of high usage, health deterioration, or charge-cycle characteristics can be flagged for further technical inspection.

The observed relationships should be interpreted as associations rather than evidence that usage directly caused changes in battery health.

7. Pricing and Partner Economics

7.1 Overview

The fifth analysis examines pricing, discounts, tariffs, revenue, and available fleet-partner information.

The objective is to understand how commercial structures are reflected in observed swap activity and revenue.

7.2 Tariff Analysis

Swap events were grouped by tariff code and compared using:

- Total swap attempts.
- Completed swaps.
- Completion rate.
- Total revenue.
- Average list price.
- Average discount.
- Average amount charged.

This provides a view of how different tariff categories contribute to network activity and revenue.

7.3 Pricing Over Time

Monthly pricing trends were analyzed using:

- Average list price.
- Average amount charged.
- Average discount.
- Total revenue.
- Total swap activity.

Examining these measures together helps distinguish changes in customer pricing from changes in overall network usage.

7.4 Fleet Partner Economics

Available fleet-partner information was combined with swap activity to examine partner-level:

- Swap volume.
- Completed swaps.
- Completion rate.

- Revenue contribution.
- Average amount charged.
- Average discount.

This provides an additional view of commercial activity across fleet relationships.

7.5 Contribution Margin

Available cost, margin, and profitability fields were reviewed to determine whether direct contribution-margin analysis could be performed from the supplied datasets.

Where a direct margin measure was available, it was considered alongside pricing and operational performance rather than evaluated independently.

7.6 Business Implications

Pricing and partner analysis can support decisions around tariff performance, discount effectiveness, partner relationships, and revenue management.

Differences between tariff or partner groups identify areas for further investigation but do not establish that a particular pricing or partner arrangement caused an observed business outcome.

8. Root Cause Analysis of Rider Retention

8.1 Overview

The sixth analysis examines rider retention and the relationship between the initial customer experience and subsequent rider activity.

The analysis focuses on whether riders returned within 30 days of their first recorded activity.

8.2 30-Day Retention

A rider-level retention dataset was created using each rider's first recorded activity date.

Riders with sufficient follow-up time were evaluated to determine whether they generated another recorded event within the following 30 days.

This provides a consistent basis for measuring early rider return behavior.

8.3 First Experience and Retention

The first recorded rider experience was compared with subsequent 30-day return behavior.

The analysis considered:

- First event outcome.
- Initial queue waiting time.
- First station used.
- Subsequent rider activity.

These comparisons help identify operational characteristics associated with different retention outcomes.

8.4 Queue Wait and Retention

Initial queue waiting time was grouped and compared with 30-day return behavior.

This provides an indication of whether differences in the initial waiting experience are associated with differences in subsequent rider activity.

8.5 Station-Level Retention

Rider retention was also compared across stations.

Only stations with sufficient rider activity were considered for meaningful comparison, reducing the influence of very small samples.

8.6 Business Implications

The analysis can help identify aspects of the first customer experience that deserve further investigation.

Stations or operating conditions associated with lower return rates can be examined alongside queue performance, service failures, and other operational metrics.

Retention differences should be treated as observational associations. They do not establish that a specific first experience directly caused a rider to return or not return.

9. Business Recommendations

The analysis provides several areas where VoltRelay Energy can focus operational and commercial improvement efforts.

9.1 Prioritize High-Failure Stations

Stations with consistently higher failure rates or lower completion rates should be prioritized for operational investigation.

Recommended actions include reviewing charger availability, battery inventory, station equipment condition, queue conditions, and recurring service issues.

9.2 Reduce Customer Waiting Time

Queue waiting time is an important customer-experience metric and should be monitored at station and time-of-day levels.

Operational teams can use hourly demand patterns and station-level queue performance to identify periods requiring additional battery availability or operational support.

9.3 Monitor Battery Health and Equipment Performance

Battery health, usage frequency, and equipment characteristics should be monitored continuously.

Batteries or equipment showing unusual performance patterns can be investigated before they contribute to service disruptions or poor customer experiences.

9.4 Use Pricing Data for Commercial Decisions

Pricing and tariff performance should be monitored together with transaction volume, revenue, and contribution margin.

Pricing changes should be evaluated using observed customer and financial outcomes rather than relying only on transaction volume.

9.5 Strengthen Rider Retention Monitoring

Rider retention should be tracked alongside the first customer experience.

Stations or operating conditions associated with lower 30-day return rates can be investigated together with queue times, failures, and other service-quality indicators.

9.6 Establish a Continuous Analytics Process

The analysis should be converted into a recurring monitoring process rather than treated as a one-time exercise.

A practical monitoring framework can track:

- Completion rate
- Failure and abandonment rates
- Queue waiting time
- Station performance
- Battery health
- Revenue and contribution margin
- Rider retention

This would allow operational teams to identify emerging issues earlier and prioritize investigations using consistent evidence.

10. Limitations

The analysis provides an evidence-based view of VoltRelay Energy's network performance, but several limitations should be considered when interpreting the results.

10.1 Observational Analysis

The analysis identifies patterns and associations in the available data. It does not establish causal relationships.

For example, an association between queue waiting time and rider retention does not by itself prove that waiting time caused a rider to leave.

10.2 Data Coverage

The analysis covers the available data for the period from January 2024 through June 2025.

Results therefore describe the observed period and may not fully represent future operating conditions.

10.3 Data Quality and Scope

Potential duplicate records and test-station activity were reviewed during data preparation. The analytical datasets were cleaned and validated before the main analysis.

Some potentially relevant external factors may not be captured in the available datasets, including broader market conditions, competitor activity, individual rider preferences, and operational events not recorded in the source data.

10.4 Retention Interpretation

The retention analysis focuses on 30-day return behavior. A rider not returning within the selected window should not automatically be interpreted as permanent customer churn.

11. Conclusion

VoltVision provides an analytical framework for understanding EV battery swapping operations across network performance, customer experience, station operations, battery health, pricing, partner economics, and rider retention.

The analysis combines transaction-level, station-level, battery-level, customer-level, and commercial data to identify operational patterns and areas requiring further investigation.

The resulting insights can support evidence-based prioritization of station improvements, customer-experience monitoring, battery and equipment management, pricing decisions, and retention initiatives.

The key value of the analysis is not simply reporting historical metrics, but establishing a repeatable approach that VoltRelay Energy can use to monitor network performance and investigate emerging operational issues over time.